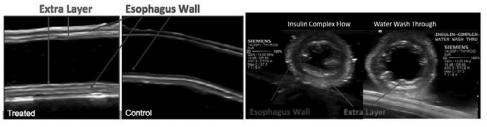
Su1827

Mbp-1 Specific Ultrasound Contrast Agents: A New Approach to Noninvasively Detect Eosinophilic Esophagitis Associated Inflammation

Hedieh Saffari, Anne Kennedy, Kathryn Peterson, Gerald J. Gleich, Leonard Pease

Background: In eosinophilic esophagitis (EoE), eosinophils invade the esophagus where they degranulate and release very cytotoxic proteins, including major basic protein (MBP-1). Currently the diagnosis of EoE necessitates endoscopy with biopsy that characterizes less than 2% of the esophagus. Furthermore, the inflammation is patchy, necessitating numerous biopsies to ensure that the involved area has been adequately sampled. Improved methods are needed to characterize the eosinophil density throughout the esophagus and to thoroughly understand this disease. Aim: To prepare the first MBP-1 specific ultrasound contrast agents to facilitate the detection of EoE associated inflammation by tethering EoE specific antibodies to their surfaces and demonstrating the binding of these agents to eosinophil proteins on ex vivo monkey esophagi. Method: The contrast agent was synthesized by preparing 2 mg/mL of insulin in HCl (pH 1.6) at 65C for 12 hours. The resulting particles were incubated with maleimide activated streptavidin. Purified antibodies specific to MBP-1 were biotinylated and reacted with the streptavidin functionalized fibers to form an insulin aggregate-steptavidin-biotin antibody complex. This complex comprises the MBP-1 specific ultrasound contrast agent. A Macaca monkey esophagus, incubated overnight in MBP-1, was filled with this solution. Ultrasound images were taken before and after washing at 15 MHz. Results: Insulin aggregates were observed using ultrasound. Incubating the insulin complex in monkey esophagi coated with MBP-1 yields an extra contrast layer at the inner wall of the esophagus, demonstrating the specific binding of these agents to eosinophil granule proteins. No such layer is visualized in control samples (not treated with MBP-1). Discussion: This is the first reported use of insulin particles as contrast agents for ultrasound. These findings indicate that insulin aggregates provide clear ultrasound contrast in liquid environments. When coupled with disease specific antibodies, these novel agents have the potential to be used for detection, minimizing the need for anesthesia or radiation associated with other diagnostic or imaging modalities. This finding is important because it provides a new avenue for clinical detection of EoE.



Longitudinal and Transverse view of control vs. treated esophagus (coated with MBP-1).

Su1828

Remission in Proton Pump Inhibitors-Responsive Esophageal Eosinophilia Correlates With Downregulation of Eotaxin-3 and TH2 Cytokines, Similarly to Eosinophilic Esophagitis After Steroids

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INTRODUCTION Proton pump inhibitor responsive esophageal eosinophilia (PPIr-Ee) is a recently described phenotype distinct from eosinophilic esophagitis (EoE). To ascertain its molecular basis may be helpful for an accurate clinical categorization of patients, as suffering from gastroesophageal reflux disease, a subphenotype of EoE, or an independent entity. AIM To address the baseline eotaxin-3 and Th2 cytokine expression profile in patients an EoE phenotype and to characterize their course after systematic PPI therapy, comparing PPI responders to EoE patients (non-responders to PPI) after topical steroids. METHODS In consecutive adult patients with dysphagia/food impaction and esophageal eosinophilia > 15 eo/HPF, eotaxin-3, IL-13 and IL-5 were tested in proximal and distal esophageal biopsies, at baseline and after PPI therapy for 8 weeks (omeprazole 40 mg twice daily). Disease remission was defined as: complete < 5 eo/HPF and partial 5-14 eo/HPF. EoE patients (unresponsive to PPI therapy) were offered rescue therapy with topical steroids for 8 weeks. RESULTS: 44 patients were included and reevaluated on PPI therapy. 20/44 patients (40%) had PPIr-Ee (12/44 (27%) complete and 8/44 (13%) partial). 14/24 EoE patients were given steroids. No differences were observed between PPIr-Ee and EoE regarding demographics, symptoms or endoscopic findings. Baseline esophageal eosinophilia and molecular characteristics were indistinguishable between PPIr-Ee and EoE, excepting marginally significant higher levels of IL-5 in proximal esophagus for EoE (Table 1). Eotaxin-3 and cytokines levels were higher in distal esophagus in PPIr-Ee, whereas were higher in proximal esophagus for EoE, excepting IL-13 (Table 1). PPI-response resulted in a significant decrease in eotaxin-3 (Figure 1), IL-5 and IL-13 levels in both distal and proximal esophagus, similar to that demonstrated after topical steroids in EoE patients (Figure 1). Higher eotaxin-3 levels were found in PPI-partial responders at baseline (distal esophagus, 1252 vs. 404, p 0.015) and after PPI therapy in distal (44 vs. 275, p 0.007) and proximal (25 vs. 168, p 0.007) esophagus, when compared to PPI-complete responders. CONCLUSIONS: PPIr-Ee and EoE remained nearly indistinguishable in most molecular parameters studied. At baseline, eotaxin-3 and Th2 cytokines expression was higher in distal esophagus for PPIr-Ee and in proximal esophagus for EoE. Histological remission in PPIr-Ee was correlated with downregulation of eotaxin-3 and Th2 cytokines, similarly to remission in EoE after topical steroids.

	PPIr-Ee Distal esophagus*	PPIr-Ee Proximal esophagus**	EoE Distal esophagus*	EoE Proximal esophagus**	p
eo/HPF, median (range)	55 (16-115)	44 (10-127)	67 (25-105)	61 (19-165)	* 0.29 ** 0.14
Eotaxin-3, median (range) relative units x 103	722 (60-2570)	462 (7-1580)	549 (13-1620)	848 (34-3490)	* 0.44 ** 0.18
IL-13 median (range) relative units x 103	18.44 (1.1-98)	12.85 (1.05-63)	15.22 (0.9-47)	14.76 (0.7-55)	* 0.87 ** 0.74
IL-5 median (range) relative units x 103	22.62 (0-50)	15.19 (0-68)	27.95 (0-143)	50.9 (4-300)	* 0.56 ** 0.09

Table 1. Baseline eosinophil peak count and molecular analysis in PPIr-Ee an EoE patients, in distal and proximal esophagus.

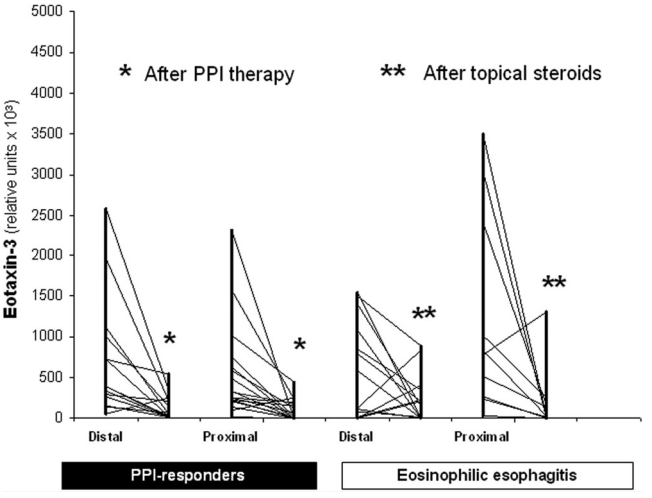

Figure 1. Eotaxin-3 levels (distal and proximal esophagus) after PPI therapy in PPIr-Ee and after topical steroids in EoE.

Su1829

Mutations in TGFβR1 and 2 Are Associated With Food Allergy, Eosinophilic Gastrointestinal Disease, and Inflammatory Bowel Disease

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Background: Loeys-Dietz syndrome (LDS) is an autosomal-dominant aortic aneurysm syndrome caused by mutations in subunit 1 or 2 of the TGF β receptor (LDS Type 1 & 2), Smad3 (LDS Type 3), or TGFβ2 haploinsuffiency (Type 4). In this study, we recognized and mechanistically characterized a strong predisposition to develop TH2 inflammation and allergic disease in LDS, including eosinophilic gastrointestinal disease. Additionally, we found that LDS patients are prone to the development of inflammatory bowel disease (IBD). Methods: The occurrence of allergic phenotypes in 58 patients with LDS Types 1 & 2 was determined using a detailed questionnaire, clinical examination, serologic testing, tissue biopsy and/or review of medical records. Peripheral eosinophil counts, plasma cytokine levels, levels of phosphorylated Smad2 (pSmad2) in thymic tissue and peripheral lymphocytes, Naïve T cell skewing patterns, and Treg frequency and function were evaluated. Patients with EGID or IBD were genotyped for common SNPs known to be associated with an increased risk of IBD. Results: Food allergy was diagnosed in 31%, eosinophilic gastrointestinal disease (EGID) in 10%, and inflammatory bowel disease in 5%. LDS patients had Th2 diseases in greater prevalence than the general population (asthma in 45%, allergic rhinitis in 48%, and eczema in 38%). LDS patients had elevated IgE levels, eosinophil counts, and TH2 cytokines (IL-5, IL-13) in their serum. LDS patients demonstrated high levels of CD4+ T cells that expressed both Foxp3 and IL-13, but retained the ability to suppress effector T cell proliferation. Naive CD4+ T cells from LDS patients differentiated into TH2-cytokine producing cells in a TGFβ-dependent manner in vitro. Levels of pSmad2 were elevated in thymi and peripheral CD4+ lymphocytes from LDS patients. Conclusion: LDS is the first Mendelian disorder to be associated with the development of food allergy, EGID, and other allergic disease. Further LDS confirms the importance of the TGF β canonical pathway in the development of IBD. These data will inform efforts to elucidate pathogenesis in nonsyndromic presentations of allergic disease and IBD and potentially reveal novel therapeutic strategies.

Su1830

T-Lymphocyte Isolation and Phenotyping From Esophageal Biopsies in Children With Eosinophilic Esophagitis: Pilot Study

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Background: Eosinophilic Esophagitis (EoE) is an immune mediated allergic disease associated with infiltration of eosinophils within the esophageal mucosa. It is believed that an inflammatory cascade involving cytokines released by T lymphocytes, in response to exposure

to food antigens and/or environmental triggers, leads to the recruitment and infiltration of eosinophils in the esophageal mucosa. Aim: To characterize T cell responses in EoE in children. Methods: We enrolled 24 patients (mean age 14.1 years, range 10-17 years, 75% male), including 2 with newly diagnosed EoE, 11 with treated EoE (3 with active inflammation >15 eos/hpf, 8 in remission <5 eos/hpf), 9 with normal endoscopy, 1 patient with mild reflux esophagitis (<5 eos/hpf) and 1 patient with Barrett's esophagus. We obtained 4 esophageal pinch biopsies at the time of clinically indicated endoscopy. For T cell isolation the biopsies were enzymatically digested. Single cell suspensions were analyzed by flow cytometry using specific cell markers for TCRαβ, CD3+, CD4+, and CD8+ T lymphocytes to phenotype the cells. A subset of cells were stimulated with PMA/ionomycin to measure intracellular cytokine potential (TNF-α and IFN-γ) via flow cytometry. Results: We extracted an average of 2.7x105 cells per patient (range 0.4to 6.5 x105). The average number of CD3 TCRαβ+ cells was 57% (range 3.9-82.9%) with 41% (range 11.3-66%) CD4+ and 44% (range 15.6-81.3%) CD8+. There was a trend towards a higher percentage of CD8+ T cells in EoE patients (n = 13) compared to the other groups (n = 11) that included normal controls and reflux patients (48% vs. 41%), although, this was not significant. The CD8+/CD4+ ratio was also slightly higher in the EoE group compared to the other groups (1.87 vs. 1.25), although these results were not significant. In 20/24 patients (11 EoE and 11 others), on average, 29% of the CD3+ cells, 32.8% of the CD4+ cells and 41% of the CD8+ cells produced TNF-α when stimulated with PMA/ionomycin (n = 20). The average IFN- γ production (n = 20) was 19.5% of CD3+ cells, 15% of CD4+ cells and 37% of the CD8+ cells. There was no significant difference in TNF- α and IFN-γ production between EoE patients and the other groups. Conclusion: We demonstrated the feasibility of isolating CD3+ TCRαβ+ T lymphocytes from pediatric esophageal biopsies and the ability to phenotype and functionally test these lymphocytes. We have observed higher CD8+ T cells in EoE patients compared to other groups and a higher CD8+/CD4+ ratio in EoE patients compared to the other groups. We also observed lower TNF- α and IFN-γ production in EoE patients compared to the other groups. Further studies are needed to determine the role of these lymphocytes in the development of EoE.

Figure 1: Flow cytometric analysis of lymphocytes extracted from biopsies.

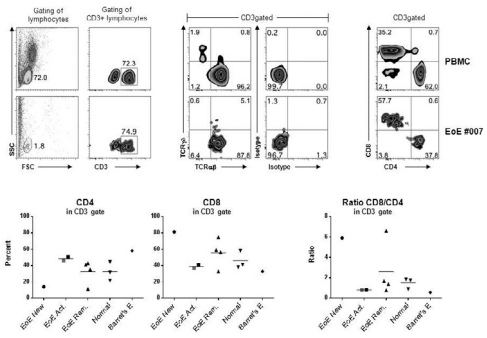
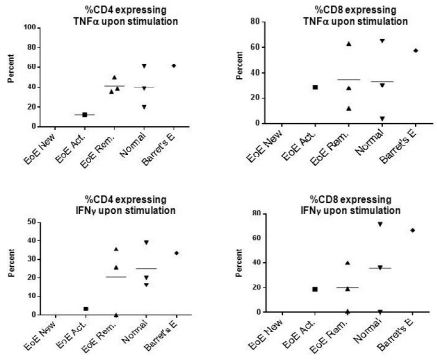


Figure 2: Flow cytometric analysis of cytokines expressed by lymphocytes.



Su1831

Determination of the Molecular Phenotype of Esophageal Mucosal Inflammation in Children With Eosinophilic Esophagitis Using a 1-Hour Esophageal String Test (EST)

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Background: Eosinophilic Esophagitis (EoE) is a chronic disease characterized by eosinophil-predominant esophageal inflammation for which endoscopy with biopsy is currently the standard method to assess mucosal inflammation and follow resolution with treatment. However, we recently reported that eosinophil secreted granule proteins (ESGPs) can be captured and assayed from esophageal secretions, and used to distinguish active EoE from EoE in remission, GERD and normal using a minimally invasive Esophageal String Test (EST) after leaving it in situ for 12-hours (1). Aims: Our objectives were to determine whether esophageal luminal biomarkers associated with esophageal eosinophilia could be captured by the EST in a shorter 1-hour sampling period and differentiate the activity of mucosal inflammation in children with EoE. Methods: Four hours before endoscopy with biopsy was performed, an EST was swallowed. After a 1-hour sampling period, the EST was removed and endoscopy was performed 3 hours later. ESGPs [major basic protein-1 (MBP1) and eosinophil peroxidase (EPX)], and Th2 cytokines (IL-4, IL-5, IL-13) and chemokines (Eotaxin-2, -3) were measured by ELISA in luminal effluents (secretions, cells) eluted from the ESTs and in extracts of esophageal biopsies obtained at the time of endoscopy (1). Results: Samples from 21 children (ages 9-18) with active EoE (n=5; 19-120 eos/HPF), treated EoE in remission (n=5; 0-9 eos/HPF), GERD (n=3; 0-2 eos/HPF) and normal esophagus (n= 8; 0-1 eos/HPF) were analyzed. EST measurements of MBP1 significantly differentiated between children with active EoE (8.07±3.6μg/ml), EoE in remission (0.59±0.17μg/ml), GERD (0.36±0.01μg/ml) and normal esophagus (0.04±0.02μg/ml) (all $p<0.01$), while EPX distinguished between active EoE (0.97±0.48μg/ml), GERD (0.05±0.02μg/ml) and normal esophagus (0.07±0.02μg/ml) (all $p<0.01$). EST measurements of eotaxin-2 differentiated between active EoE (756±356pg/ml) and treated EoE in remission (82±24pg/ml) ($p<0.05$) and normal (99.7±15pg/ml, p=0.052), while Eotaxin-3 differentiated between active EoE (442±200pg/ml) and normal esophagus (2.5±2.5pg/ml) ($p<0.05$). Th2 cytokines (IL-4, IL-5, IL-13) were not detectable in 1-hour EST samples (all below sensitivity of the ELISAs). EST MBP1 (r=0.972, $p< 0.001$) and EPX (r=0.751, $p<0.001$) biomarker levels correlated significantly with the peak number of eosinophils in biopsies, as did EST levels of Eotaxin-2 (r= 0.917, $p<0.001$) and Eotaxin-3 (r=0.917, $p<0.001$). Conclusions: Eosinophil-associated biomarkers that are captured by the EST in a clinically relevant time frame reflect mucosal inflammation and disease activity in EoE. The EST is a novel, time efficient, minimally invasive device for measuring esophageal eosinophilic inflammation in children with EoE. Reference: (1) Furuta GT et al. Gut. Aug. 15, 2012; PMID: 22895393.

Su1832

Duration of Untreated Inflammation Represents the Main Risk Factor for Stricture Development in Eosinophilic Esophagitis

Alain Schoepfer, Ekaterina Safroneeva, Christian Bussmann, Tanja Kuchen, Susanne Portmann, Alex Straumann

Background: Eosinophilic Esophagitis (EoE) is a chronic-inflammatory condition presenting either as inflammatory-phenotype (IP), as stenosing-phenotype (SP) or as an overlap form. The IP is endoscopically characterized by whitish exudates, furrows and edema, whereas the SP shows signs reflecting tissue remodeling, such as rings and strictures. There is some evidence that SP develops over time, but the precise evolution of stricture formation during the disease course remains largely unknown. Aim: To correlate the prevalence of esophageal strictures with the duration of untreated disease and to evaluate risk factors for stricture formation. Methods: Retrospective analysis of the Swiss EoE Database (SEED), extended by a review of patients charts, endoscopy and pathology records. Strictures were assessed endoscopically at first diagnosis of EoE and defined as a narrowing leading to problems passing a standard adult upper endoscope (outer diameter 9mm). Diagnostic delay was defined as period from onset of EoE symptoms to diagnosis. Results: Two hundred EoE patients were analyzed (153 males, mean age at index visit 39 ± 15 years, all Caucasians). All patients were symptomatic; dysphagia was present in 94% and chest pain in 35% of patients. Allergies were identified in 132 (66%) of patients. An endoscopic bolus removal was performed in 56 patients (28%), whereof 28 prior and 28 at the time of EoE diagnosis. Median diagnostic delay was 6 years (IQR 2-12, range 0-36 years). We observed the following stricture prevalence at EoE diagnosis according to the following intervals of disease duration: 0-2 years (n=58) 17.2%, 3-5 years (n=39) 30.8%, 6-8 years (n=18) 38.9%, 9-11 years (n= 29) 37.9%, 12-14 years (n=12) 41.7%, 15-17 years (n=14) 64,3%, 18-20 (n=6) 66.7%, >20 years (n=24) 70.8% ($p < 0.001$, Cochrane-Armitage trend test). Logistic regression modeling detected a long duration of untreated disease (defined as diagnostic delay ≥ 7 years) to be strongly associated with the presence of strictures at the index visit (Odds Ratio 3.24, 95% confidence interval 1.75-5.97, $p < 0.001$). No association of strictures at the index visit was found with gender, presence of allergies, age at EoE diagnosis, positive family history of EoE, blood eosinophilia, elevated serum IgE, and peak eosinophil count at index histology. Conclusions: Strictures in EoE develop over time. The frequency of stricture formation is proportional to the duration of the untreated disease. These findings underscore the need to reduce the diagnostic delay in EoE and to control the underlying inflammation in order to prevent esophageal remodeling.

Su1833

Eosinophilic Esophagitis Is a Progressive Fibrostenotic Disease: Insights From a Phenotypic Analysis

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Background: There is limited information about the natural history of eosinophilic esophagitis (EoE), and it is unknown whether specific phenotypes of EoE are the result of disease progression. **Aim:** To describe the clinical features of EoE patients with inflammatory, fibrostenotic, and mixed phenotypes, determine predictors of these phenotypes, and make inferences about the natural history of EoE. **Methods:** This was a retrospective study of the University of North Carolina EoE Clinicopathologic database from 2001-2011. Subjects with an incident diagnosis of EoE who met consensus guidelines were included. All had symptoms of esophageal dysfunction, ≥15 eos/hpf (hpf area=0.24 mm^2), and did not respond to a PPI trial. The phenotypes were defined as fibrostenotic if there were esophageal rings, narrowing, or strictures and no evidence of linear furrows or white plaques; as inflammatory if there were furrows, plaques, or a normal esophagus and no evidence of fibrostenotic changes; and as mixed if there were a combination of findings. Clinical and histologic features were compared between the three phenotypes. Multinominal logistic regression was performed to assess predictors of phenotype status. **Results:** Of 374 EoE cases, 134 (36%) were inflammatory, 163 (43%) were mixed, and 77 (21%) were fibrostenotic. Those with an inflammatory phenotype were more likely to be younger than those with mixed or fibrostenotic phenotypes (13 vs 29 vs 39 years, respectively; $p <0.001$). Dysphagia was less likely in inflammatory compared to mixed or fibrostenotic (36% vs 77% vs 92%; $p <0.001$), as was food impaction (15% vs 37% vs 39%) and esophageal dilation (0% vs 24% vs 47%; $p <0.001$). Abdominal pain, vomiting, and failure-to-thrive were more common in inflammatory ($p<0.001$ for all), as were atopy and food allergies ($p <0.05$). The mean symptom length prior to diagnosis was shorter for inflammatory compared to mixed and fibrostenotic (5 vs